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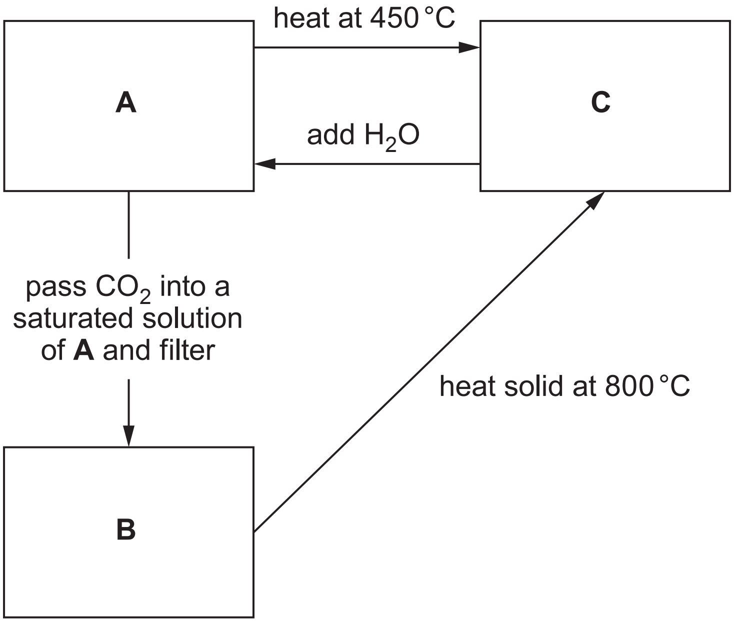
12. *(a)* A series of experiments is performed on three white solids, **A**, **B** and **C**. All contain the same Group 2 metal ion. Compound **A** is the metal hydroxide.

The experiment is summarised below.



- The compounds give a definite colour in a flame test
- Compound **A** is only slightly soluble in water
- Compound **B** is insoluble in water

(i) A student correctly identified the metal ion as calcium. Give **two** reasons why she came to that conclusion. [2]

....

....

....

....

(ii) Name compound **B**. [1]

....

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(iii) 0.110 g of compound **C** was completely neutralised by 26.10 cm^3 of 0.150 mol dm^{-3} hydrochloric acid.

Given that 1 mol of compound **C** reacts with 2 mol of acid, calculate the relative formula mass of compound **C** and hence confirm that the metal ion is calcium.

You **must** show your working. [2]

(b) Group 2 metals are not the only elements that form 2+ ions.

A sample of an element has two isotopes, one with 70 neutrons and the other 72 neutrons. This element forms a 2+ ion containing 48 electrons. The relative abundances of the isotopes are 57.9 % and 42.1 % respectively.

Calculate the relative atomic mass of the element. Give your answer to **four** significant figures. [3]

A_r = ...

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(c) Describe, giving **brief** experimental details, how you would obtain the maximum amount of magnesium carbonate from a known mass of magnesium, in a **two-stage** process. Include appropriate chemical / ionic equations in your answer. [6 QER]

14

